

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which organization handles e-waste recycling in India?
 - a) CPCB
 - b) ISRO
 - c) RBI
 - d) FSSAI
- 2 Which country is the largest producer of e-waste?
 - a) India
 - b) USA
 - c) China
 - d) Brazil
- 3 What is the full form ofEPR in e-waste management?
 - a) Environmental Product Reuse
 - b) Extended Producer Responsibility
 - c) Electronic Public Recycling
 - d) Eco-friendly Product Rule
- 4 Improper e-waste disposal causes—
 - a) Soil pollution
 - b) Water pollution
 - c) Air pollution
 - d) All of the above
- 5 What is the first step in e-waste management?
 - a) Burning waste
 - b) Collection
 - c) Landfilling
 - d) Mixing with household waste

- 6 E-waste collection bins usually display—
 - a) Skull sign
 - b) Crossed-out wheeled bin symbol
 - c) Recycle triangle only
 - d) Biohazard symbol
- 7 Which metal is commonly recovered from e-waste?
 - a) Aluminium
 - b) Gold
 - c) Sodium
 - d) Calcium
- 8 Which of the following is mainly responsible for e-waste generation?
 - a) Increase in gadget use
 - b) Agriculture
 - c) Fishing
 - d) Textile industry
- 9 Refurbishing electronics means—
 - a) Burning old devices
 - b) Repairing and making them reusable
 - c) Breaking them into scrap
 - d) Mixing with household waste
- 10 Hydrometallurgy involves—
 - a) Use of high temperature
 - b) Use of chemical solutions for extraction
 - c) Burning material
 - d) Manual separation
- 11 Which type of recycling releases harmful fumes if done informally?
 - a) Eco-friendly recycling
 - b) Open-wire burning
 - c) Chemical leaching
 - d) Dismantling
- 12 The first step in HRA is—
 - a) Exposure assessment
 - b) Hazard identification
 - c) Risk communication
 - d) Risk management

Q2. Solve the following.

- A) Explain the major sources and categories of electronic waste with suitable examples. 6
- B) Define the environmental impacts of improper E-waste disposal? Explain its effects on air, water, and soil. 6

Q3. Solve the following.

- A) Explain the concept of health risk assessment. Describe its importance in evaluating environmental hazards. 6
- B) Explain how technological innovations have improved the recycling of electronic waste. 6

Q4. Solve Any Two of the following.

- A) Explain the major challenges in managing e-waste in urban areas. 6
- B) Describe the major hazardous substances released during informal E-waste recycling and explain their health impacts. 6
- C) Explain how improper metal disposal from electronic waste affects the environment. 6

Q5. Solve Any Two of the following.

- A) Explain the dangerous elements found in electronic waste. 6
- B) Explain the process of electronic waste recycling with a neat flow diagram. 6
- C) Describe the health risks to humans due to improper handling and disposal of electronic waste. 6

Q6. Solve Any Two of the following.

- A) Explain the contribution of material recovery to circular economy principles and long-term resource sustainability. 6
- B) Explain the role of government, industry, and consumers in effective and responsible E-waste management. 6
- C) State the rules and regulation for electronic waste prescribed by Govt. of India. 6

*** End ***